

STUDENT INFORMATION

Name:

Date:

VEHICLE APPLICATION

Make	Model	Year	Engine
Transmission	Production Date	Model Code	Mileage

Section A - Vehicle Prep and Initial Measurements

- 1 Verify the battery is healthy before disconnecting any electrical or doing brake work that may require pump cycling.

Battery OCV (measured volts)	Spec (12.6V fully charged)	OK / NG
	12.6V	

- 2 Remove the front wheels using the proper torque sequence in reverse. Place wheels safely.

Wheel lug torque spec from ShopKey (ft-lb):

- 3 Inspect the brake pads in place. Note thickness and any uneven wear, glazing, or contamination.

- 4 Measure rotor thickness at 6 points around the rotor surface (3 inboard, 3 outboard, evenly spaced).

Point 1	Point 2	Point 3	Point 4	Point 5	Point 6
Min Reading		Max Reading		Thickness Variation	

Disc Brakes Lab

Step 3 of 3 · Walkthrough

30 pts

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- 5 Mount the dial indicator on the hub and measure lateral runout. Take readings at multiple points around the rotor.

Reading 1	Reading 2	Reading 3	Reading 4	Reading 5	Max Spec	OK / NG

- 6 If runout is out of spec, perform rotor matching. Record runout reading at each of 5 stud positions.

Stud Position	Runout Reading	OK / NG
Position 1 (original)		
Position 2		
Position 3		
Position 4		
Position 5		

- 7 Remove the caliper from the rotor. Inspect for fluid leaks, damage to the dust boot, and damage to the caliper body.

- 8 Remove the caliper bracket. Pull both slide pins. Inspect bushings, pin condition, and dust boots. Note any corrosion, swelling, or seizure.

SIGN-OFF - SECTION A COMPLETE (INITIAL INSPECTION)

Instructor Initials:

Date:

Section B - On-Car Brake Lathe Refinishing**BEFORE YOU CUT**

Verify the rotor is above the machine-to spec. Have your instructor verify the lathe setup before cutting.
Refinishing below the machine-to spec means the rotor must be replaced, not cut.

- 9 Open the caliper bleeder screw. Retract the piston using a C-clamp or piston retraction tool. Catch fluid in a drain bottle. **CLOSE THE BLEEDER** when done.

- 10 If the rear caliper has an integrated parking brake, retract the piston with the proper twist tool (clockwise rotation while pressing in). For Electric Parking Brake (EPB), use the scan tool service mode before retraction.

Retraction method used:

- 11 Remove the brake pads. Inspect the pad backing plates and abutment hardware. Note any rusted clips, missing shims, or hardware that needs replacement.

- 12 Set up the Pro-Cut on-car brake lathe. Verify all attachment points and trim adjustments. Get instructor verification **BEFORE** cutting.

Instructor initials confirming setup:

- 13 Cut both rotor faces with the Pro-Cut. Make passes until the surface is clean, with even hatch pattern across the friction surface.

- 14 Re-measure rotor thickness after cutting. Compare to MIN THICKNESS spec.

Final Thickness After Cut	MIN Spec	Above Min - OK / NG

SIGN-OFF - SECTION B COMPLETE (ON-CAR LATHE DONE)

Instructor Initials:

Date:

Section C - Off-Car (Bench) Lathe Refinishing

- 15 Locate a rotor in the machine room for off-car lathe work. Note the rotor application if marked.

Rotor application or part number (if marked):

- 16 Measure the initial thickness of the rotor at 6 points and find the minimum. Look for cast or stamped MIN THICKNESS spec on the rotor.

Point 1	Point 2	Point 3	Point 4	Point 5	Point 6

Initial Minimum Reading

MIN Spec on Rotor

- 17 Set up the off-car bench lathe. Mount the rotor on the proper arbor, install silencer band, and zero the cutting head.

Instructor initials confirming setup:

- 18 Make the cut. Cut both faces in a single pass to maintain parallelism.

- 19 Measure final rotor thickness after the cut. Compare to MIN THICKNESS spec.

Final Thickness After Cut	MIN Spec	Above Min - OK / NG

SIGN-OFF - SECTION C COMPLETE (OFF-CAR LATHE DONE)

Instructor Initials:

Date:

Section D - Reassembly and Burnish

- 20 Clean the caliper mounting bracket and abutment surfaces. Remove all rust and old grease.
- 21 Lubricate the slide pins with silicone synthetic brake lubricant **ONLY**. Do not use moly or chassis grease. Reinstall pins.
- 22 Install the mounting bracket. Torque to spec.
- Mounting bracket torque spec from ShopKey (ft-lb):
- 23 Install new brake pads. Verify shim and clip placement matches the original. Apply a thin film of brake lube to abutment contact points only.
- 24 Install the caliper. Torque the slide bolts to spec.
- Caliper slide bolt torque spec from ShopKey (ft-lb):
- 25 Reinstall the wheels using the proper torque sequence. Pump the brake pedal until firm **BEFORE** moving the vehicle.
- 26 Document any TSBs you searched for or found related to your work today.

SEARCH TERMS USED**TSBs FOUND - LIST NUMBER AND TITLE****SIGN-OFF - SECTION D COMPLETE (LAB FINISHED)**

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Date:

Disc Brakes Lab

Step 3 of 3 · Walkthrough

30 pts

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GRADE BREAKDOWN

QUESTION / SECTION	WHAT EARNS FULL CREDIT	POINTS
Battery OCV test	Voltage measured and OK/NG marked	1
Wheel torque + pad inspection	Spec recorded and pads inspected with notes	1
Rotor thickness measurements (6 pts)	All 6 measurements recorded with min, max, variation	3
Lateral runout measurements	5 readings recorded with spec and OK/NG	2
Rotor matching procedure	All 5 stud positions measured with OK/NG	2
Caliper inspection	Leak, dust boot, and damage condition documented	1
Slide pin inspection	Bushings, pin condition, dust boots all noted	1
Sign-off Section A complete	Instructor initialed the completed section	2
On-car lathe setup verification	Instructor signature on setup	1
Pad and hardware findings	Backing plate, clip, and shim condition documented	1
On-car final thickness vs spec	Measured final thickness compared to MIN spec	2
Sign-off Section B (on-car lathe)	Instructor initialed the completed section	2
Off-car initial measurements	6 thickness points + min reading + spec recorded	2
Off-car lathe setup verification	Instructor signature on setup	1
Off-car final thickness vs spec	Measured final thickness compared to MIN spec	2
Sign-off Section C (off-car lathe)	Instructor initialed the completed section	2
Reassembly torques	Bracket and caliper torques recorded from ShopKey	1
TSB documentation	Search terms and TSB findings documented	1
Sign-off Section D (final)	Instructor initialed the completed section	2
TOTAL - STEP 3: WALKTHROUGH		30 pts

Reminder: All four instructor sign-offs must be initialed before uploading this PDF to Canvas.